Simple-Structure Fixed-Frequency Isolated Resonant Converter for EV Battery Charger Applications With CC–CV Charge Method

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***Abstract*—This article presents a simple fixed-frequency reconfigurable resonant converter (RC) with constant-current (CC)—constant-voltage (CV) charge for electric vehicle batteries. The CC–CV charge is implemented using the LCC converter inherent CC characteristic and the series LC converter inherent CV characteristic when operating at the resonant frequency. The proposed battery charger operating modes from CC to CV is performed by a dual resonant network, which is reconfigured by an auxiliary switch resulting in the battery charging time reduction and efficiency improvement. The output current variation of the conventional method in CC mode is minimized using the auxiliary switch to block the feed-forward path, while it provides better efficiency by operating as a synchronous rectifier (SR) in CV mode. The proposed converter operation and performance is analyzed in detail. An 180 W experimental prototype generating a 50–90 V output voltage and 0.2–2 A output current from a 340 V input voltage is designed and implemented to confirm the theoretical analysis. The maximum efficiency of 98.8% is achieved due to the structure simplicity, using the SR at CV mode, operating at the resonance frequency, zero voltage switching operation of the switches, and zero current switching operation of the diodes.**

***Index Terms*—Battery charger, CC–CV charge, dual resonant network, zero-current switching (ZCS), zero-voltage switching (ZVS).**

I. Introduction

FOSSIL fuels have played a dominant role in energy consumption of transportation and industries for centuries. However, fossil fuels combustion causes greenhouse gas emissions and environmental pollution. These issues have turned attention toward electric vehicles (EVs) with the advantage of being eco-friendly [1], [2], [3], [4]. Battery is one of the vital elements of EVs and lithium-ion batteries are known as the most suitable choice to be used in EVs due to their high power density, long life, enhanced safety, and fast charging times [5], [6]. The commonly used charging method for lithium-ion batteries, is the constant current–constant voltage (CC–CV) charge method due to its fast charging and versatility. In addition to the above mentioned advantageous, the CC–CV charging contributes to longer battery life and enhanced safety by switching between CC charging to avoid over-current charging and CV charging to prevent over-voltage [7], [8]. The CC–CV charging method includes two distinct operating modes. In the first mode, the battery is initially charged using a constant current (CC) to a fixed upper limit voltage. In the second mode, the battery is charged with a constant voltage (CV) while the charging current starts to decrease until it drops to a preset cut-off current value [9]. The battery charger is another fundamental component of the EVs, and improving its specifications plays a fundamental role in the development of EVs [10], [11], [12]. Resonant converters (RCs) have emerged as appropriate options in the EVs battery chargers due to their high efficiency, soft-switching capability, and compact size [13], [14], [15]. However, battery charging with the CC-CV method needs wide changes in the battery charger output voltage, and RCs are required to operate in a wide frequency range if they are used in their conventional operation. To restrict the RCs operating frequency range for a wide voltage gain range, many methods are presented. The most common method is integrating a basic PWM converter at the RC primary [16], [17], [18], [19]. As a result, both the resonant network pulse width of the input square voltage and its amplitude depend on the inverter switches duty cycle. Therefore, the converter voltage gain dependence on the duty cycle variations increases and the converter can be controlled by pulse width modulation (PWM) control. However, in this method, due to the cascade structure, the power path elements and as a result the conduction losses increase. In another common method, the RC is designed as a reconfigurable structure [17], [20]. In this method, by changing the configuration of the converter, several separate operational modes are created, in which a specific voltage gain range is provided in each operating mode. The set of these modes leads to the production of a wide voltage gain and the RC can be controlled as PWM or by limited frequency variation. However, this increases the converter structure complexity and control. In some modified RCs, adjustable turns ratio transformers are used in order to reduce the frequency range [21]. In these converters, by changing the transformer turns ratio, a wide voltage gain range is achieved. However, this method increases the converter volume and causes complexity of structure and control. RCs exhibit an inherent characteristic of load-independent output voltage or current when operating at

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their resonant frequencies, and this feature can be used to constraint the frequency range. In some modified RCs, this inherent characteristic is used to provide both constant output voltage and constant output current modes required to provide CC–CV battery charge [22], [23], [24]. In these structures, the mode change from CC to CV is performed via changing a variable resonant inductor or capacitor to change the resonant frequency or by reconfiguring multiple resonant networks. However, the number of elements and thus the converter complexity increases significantly. In this article, a novel converter with a dual resonant network is proposed which can be employed as an EV battery charger having CC–CV charge capability with limited frequency variation. The proposed converter, through its innovative topology, implements the CC–CV battery charging requirement of wide voltage gain variation using a simple fixed frequency control, unlike conventional RCs which require a sophisticated control with wide switching frequency variations to provide such a wide gain, causing significant issues. The frequency variation is constraint due to the reconfigurable structure in two forms of LCC RC and LC series RC. The proposed converter employs dual resonant network with the preserved inherent LCC RC constant output current characteristic due to using an auxiliary switch which blocks the feed–forward path during CC operation while, in CV mode, the resonant network configuration alters to a series LC RC providing a constant output voltage. Therefore, the possibility of accurate operation in two separate modes of CC and CV is provided using an auxiliary switch which can reduce the battery charging time. It is worth noting that in the proposed battery charger, owing to the novel design, unlike similar resonant battery chargers referenced in [9], [22], [24], which leverage the inherent constant output current and constant output voltage characteristics during resonant frequency operation for CC–CV battery charging, a single MOSFET auxiliary switch, which is used to switch from CC to CV mode, is employed instead of bidirectional switches. Bidirectional switches often pose issues such as low quality and complex drive requirements. By using a single MOSFET, the proposed design effectively avoids the problems associated with bidirectional switches, ensuring a smoother transition from CC mode to CV mode. Additionally, the proposed converter has a simpler structure, and fewer components compared with the mentioned topologies. Furthermore, in this converter, the auxiliary switch is employed as a synchronous rectifier to considerably reduce the power loss during the CV mode operation. The proposed converter operates at the resonant frequency, which minimizes circulating power loss and improves efficiency. Zero-voltage switching (ZVS) of the switches and zero-current switching (ZCS) of the diodes are achieved during both CC and CV modes of operation. Additionally, compared with other solutions such as the LLC converter which operates in two resonant frequencies (series and parallel) to provide CC and CV modes without frequency change, where ZVS of inverter switches is lost during parallel resonant frequency mode, the proposed solution maintains ZVS in both modes.

II. Proposed Battery Charger Operation Principles

In the proposed battery charger shown in Fig. 1, two separate functions of the CC and CV required for EV battery are provided using the RCs inherent characteristics when operating at the resonant frequency. A dual resonant network used in this RC reconfigures between LCC and series LC networks through keeping an auxiliary switch on or operating it as a synchronous switch. Thus, when CC charge is required, the converter produces a constant output current by operating as LCC RC [Fig. 2(a)] at its resonant frequency. Similarly, when CV charge is required, the converter operates as a resonant series LC converter [Fig. 2(b)] and produces a constant output voltage by operating at the series LC resonant frequency. In this way, CC–CV charge is provided for the EV battery without any need for extensive frequency variations.

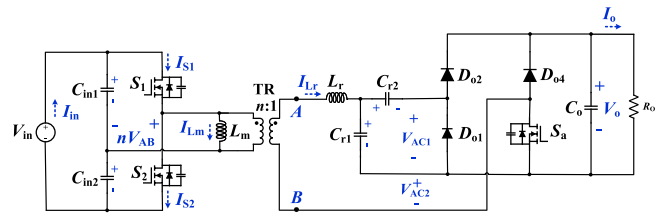

Fig. 1. Proposed battery charger.

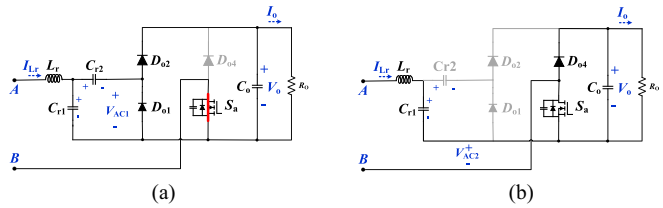

Fig. 2. Equivalent circuit of the proposed battery charger at the transformer secondary side: (a) CC mode; and (b) CV mode.

The proposed battery charger is structured based on a dual resonant network (resonant capacitors C_{r1} and C_{r2} and resonant inductor L_r), main switches (S_1 and S_2), two half-wave rectifiers (one half-wave rectifier includes diodes D_{o1} and D_{o2} and another includes diode D_{o3} and the auxiliary switch S_a), transformer (TR), input capacitors ($C_{\text{in}1}$ and $C_{\text{in}2}$), and the output capacitor (C_o).

A. CC Mode Operation

The CC mode is realized through keeping the auxiliary switch S_a on which forces the rectifier diode D_{o3} to be reverse biased. In this mode, the half-wave rectifier including D_{o1} and D_{o2} is activated, which enables the LCC resonant network to produce the output voltage V_{AC1} as shown in Fig. 2(a). When the LCC RC operates at the resonant frequency, a constant output current independent of load variations is provided for the battery. The proposed battery charger key waveforms at CC mode are depicted in Fig. 3, where, S_1 and S_2 are the inverter switches which are gated complementary to each other. Since the proposed converter operation in each half cycle is similar, only the first half-cycle of operation is described. The converter first operating half-cycle is split into four operating intervals and the equivalent circuit of each interval is exhibited in Fig. 4(a)–4(d).

Interval 1 [t_0–t_1]: Interval 1 starts when the S_1 switch is turned on at ZVS. Thus, the voltage $V_{\text{in}}/2n$ is applied to the

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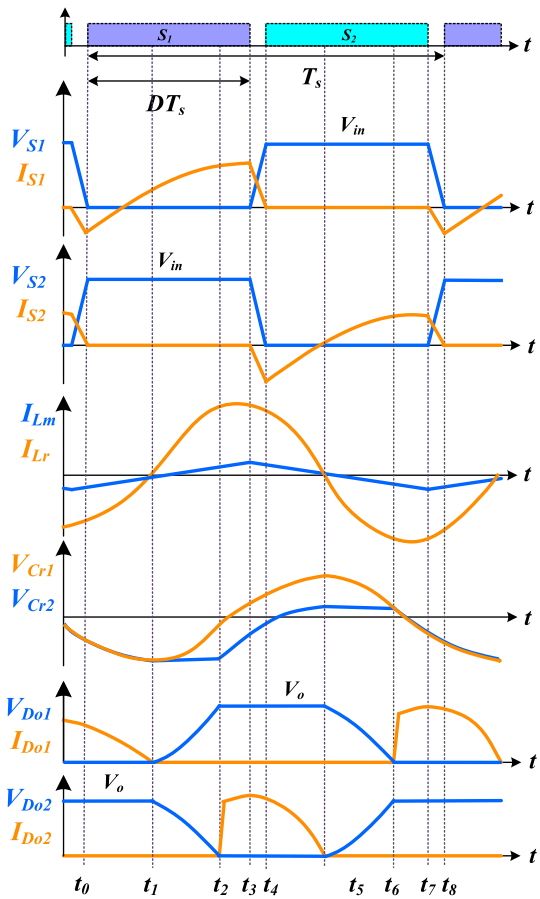

Fig. 3. Proposed battery charger key waveforms in CC charge operation.

LCC resonant network input while $V_{\text{in}}/2$ is applied to the magnetizing inductance L_m and causing it to charge. Additionally, D_{o1} diode conducts and the resonant network output V_{AC1} is short-circuited while D_{o2} is reverse biased. Through a resonance between L_r, C_{r1} and C_{r2}, the resonant inductor L_r is charged and the resonant capacitors C_{r1} and C_{r2} are discharged. Once the I_{Lr} current reaches zero, D_{o1} turns off and this interval ends.

Interval 2 [t_1–t_2]: At the beginning of this interval, D_{o1} turns off and the resonance in the previous interval stops and C_{r2} voltage remains constant until the next interval. But L_r and C_{r1} continue to resonate until the voltage summation of the resonant capacitors ($V_{Cr1} + V_{Cr2}$) reach V_o and D_{o2} turns on at t_2.

Interval 3 [t_2–t_3]: When D_{o2} starts conducting at the beginning of this interval, C_{r2} begins to participate in resonance with L_r and C_{r1} which causes it to charge while V_o is applied to the resonant network output. This Interval ends when S_1 is turned off.

Interval 4 [t_3–t_4]: By turning S_1 off at the beginning of this interval, the summation of I_{Lm} and I_{Lr}/n charge and discharge the S_1 and S_2 output capacitors, respectively. When S_2 output capacitor is fully discharged, the S_2 body diode begins to conduct and S_2 can be turned on under ZVS.

B. CV Mode Operation

Constant voltage mode occurs when the auxiliary switch S_a operates as a synchronous rectifier and the half-wave rectifier including D_{o3} and S_a is activated. In this mode, the half-wave rectifier consisting of D_{o1} and D_{o2} is disabled, which causes the capacitor C_{r2} to be inactivated. In this mode, the set of L_r and C_{r1} forms a series resonant network whose output V_{AC2} is rectified by D_{o3} and S_a. By operating the converter at the series network resonant frequency, a constant output voltage independent of load variations is provided for battery CV charge. The operating principles of this mode are not discussed in this article due to resemblance with the conventional series RC which is accessible in [25].

III. Proposed Battery Charger Analysis

A. Conversion Ratio

Due to operating the proposed converter at the resonant frequency, most of the power transferred to the output is in the fundamental harmonic form and the fundamental harmonic approximation (FHA) analysis can be carried out with satisfactory accuracy for the theoretical analysis [26], [27]. The LCC and LC resonant network ac equivalent circuits are depicted in Fig. 5(a) and 5(b), respectively. V_{AB} and $V_{AC_{1,2}}$ are asymmetric square waveform voltages applied to the resonant network input and output, respectively. Accordingly, the normalized frequency ω_n, resonant capacitors ratio C_n, characteristic impedance Z_0, quality factor $Q_{1,2}$, and the resonant network equivalent output resistance are defined as below

$$\omega_n = \frac{\omega_s}{\omega_r}, \quad C_n = \frac{C_{r2}}{C_{r1}} \tag{1}$$

$$Z_0 = \sqrt{\frac{L_r}{C_{r1}}}, \quad Q_{1,2} = \frac{Z_0}{R_{ac1,2}}, \quad R_{ac1,2} = \frac{2R_o}{\pi^2} \tag{2}$$

where R_o is the converter output resistance and $\omega_r = \omega_s = (1/\sqrt{L_r C_{r1}})$ at CC mod and CV mode. The amplitude of V_{AB} and $V_{AC_{1,2}}$ is $V_{\text{in}}/2n$ and V_o, respectively and their fundamental harmonic amplitude are calculated as follows:

$$|V_{AB}^F| = \frac{2V_{\text{in}}}{n\pi}, \qquad |V_{AC1,2}^F| = \frac{2V_o}{\pi}. \tag{3}$$

The LCC resonant network current gain and the LC voltage gain are calculated as follows:

$$H(j\omega_n) = \frac{I_{AC1}}{\frac{V_{AB}}{Z_0}} = \frac{jQ_1 C_n \omega_n}{Q_1(1-\omega_n{}^2(1+C_n)) + jC_n\omega_n(1-\omega_n{}^2)} \tag{4}$$

$$M(j\omega_n) = \frac{V_{AC2}}{V_{AB}} = \frac{1}{(1 + j\omega_n Q_2(1 - \frac{1}{\omega_n{}^2}))}. \tag{5}$$

The battery charger output current I_o at CC mode and the output voltage V_o at CV mode are obtained as follows:

$$I_o = \frac{2V_{\text{in}}}{n\pi^2 Z_0} \qquad < \text{at CC} > \tag{6}$$

$$V_o = \frac{V_{\text{in}}}{n} \qquad < \text{at CV} >. \tag{7}$$

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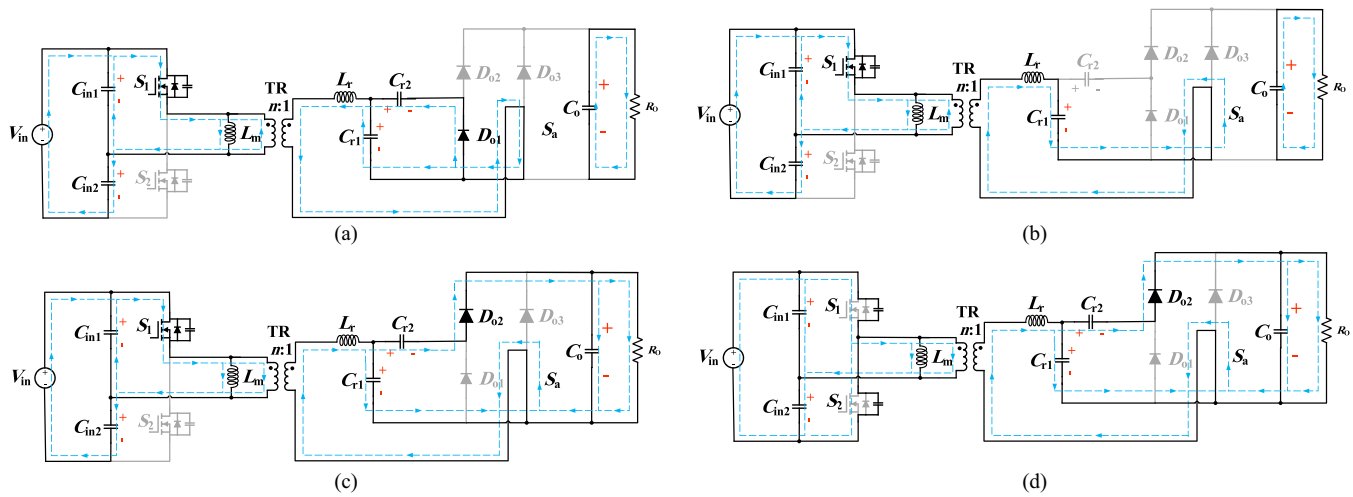

Fig. 4. Proposed battery charger CC mode operating intervals: (a) interval 1[t_0-t_1]; (b) interval 2[t_1-t_2]; (c) interval 3[t_2-t_3]; and (d) interval 4[t_3-t_4].

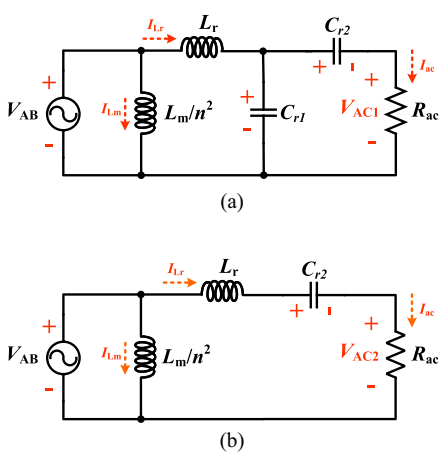

Fig. 5. Resonant network ac equivalent circuit: (a) LCC; and (b) LC.

B. Semiconductor Devices Voltage and Current Stress

The inverter switches voltage and current stress ($\widehat{V_S}$ and $\widehat{I_S}$) are calculated as follows:

$$\widehat{V_S} = V_{\text{in}}, \quad \widehat{I_S} = I_{Lr}(t_3) + \frac{\Delta I_{Lm}}{2}. \tag{8}$$

The resonant inductance current I_{Lr} and magnetic inductance current ripple ΔI_{Lm} are obtained as below

$$\Delta I_{Lm} = \frac{V_{\text{in}}}{4fL_m} \tag{9}$$

$$I_{Lr}(t) = \frac{2V_{\text{in}}}{n\pi Z_0}\sqrt{\frac{1}{Q_1^2} + \frac{(1+C_n)^2}{C_n^2}}\cos(\omega_s t - \theta) \ < \text{at CC} > \tag{10}$$

$$\theta = \tan^{-1}\left(Q_1\left(\frac{1+C_n}{C_n}\right)\right) \tag{11}$$

$$I_{Lr}(t) = \frac{2V_{\text{in}}Q_2}{n\pi Z_0}\cos(\omega_s t) \ < \text{at CV} >. \tag{12}$$

The voltage and current stress of semiconductor devices, on the transformer secondary side, defined as ($\widehat{V_D}$ and $\widehat{I_D}$) as well as their average current ($\widetilde{I_D}$), are obtained as follows:

$$\widehat{V_D} = V_o, \quad \widehat{I_D} = \pi I_o, \quad \widetilde{I_D} = I_o. \tag{13}$$

C. Soft-Switching Condition

To provide ZVS turn-ON for S_1 and S_2 switches within the entire operating range, the summation current ($I_{Lr}/n + I_{Lm}$) must be large enough to discharge and charge S_1 and S_2 output capacitors (C_{oss}), respectively at t_7. Similarly, the opposite action of charging and discharging S_1 and S_2 output capacitors (C_{oss}), respectively should be performed at t_3. The ZVS turn-ON criteria for S_1 and S_2 are expressed as follows:

$$-\left(\frac{I_{Lr}(t_7)}{n} - \frac{\Delta I_{Lm}}{2}\right) \geqslant \frac{2C_{\text{oss}}\widehat{V_S}}{t_d} \quad \text{for } S_1 \tag{14}$$

$$\left(\frac{I_{Lr}(t_3)}{n} + \frac{\Delta I_{Lm}}{2}\right) \geqslant \frac{2C_{\text{oss}}\widehat{V_S}}{t_d} \quad \text{for } S_2 \tag{15}$$

where, t_d is the deadtime between the switches gate signals. To ensure ZVS turn-ON for S_1 and S_2, (14) and (15) must be satisfied for the entire output voltage and current. Due to the weaker ZVS characteristic of the series LC converter rather than LCC network, the worst-case condition occurs at CV mode where series LC network is activated. As a result, if the ZVS turn-ON condition is provided for CV mode, it would also be provided for CC mode.

IV. Experimental Result

A 180 W experimental prototype operating at 200 kHz switching frequency for the wide output voltage range of 50–90 V and 0.2–2 A output current is realized to validate the theoretical analysis. The CC–CV charge profile of the EV battery and its equivalent impedance during the charge is depicted in Fig. 6(a). During the CC charging mode, the battery is charged by a 2 A constant-current, and the converter output voltage increases

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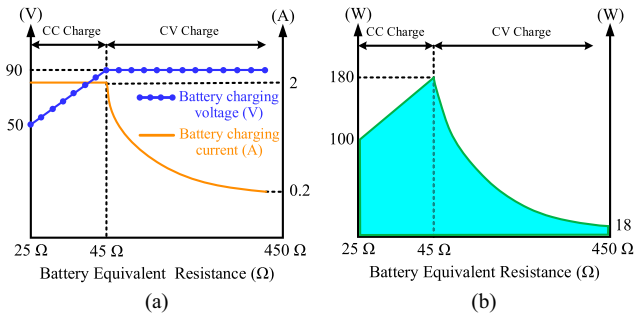

Fig. 6. (a) CC–CV charge profile of the EV battery pack and its equivalent impedance during the charge; and (b) Power distribution in CC and CV modes.

from 50 V to the fixed upper limit voltage 90 V. As soon as the CC mode ends, the CV mode starts. During CV mode, the battery is charged with the 90 V constant-voltage while the charging current starts to reduce until it drops to the preset cut-off current of 0.2 A. Also, the power distribution in CC and CV modes operation is presented in Fig. 6(b).

By utilizing (7), the transformer turns ratio is obtained as $n = 3.75$. Then, the characteristic impedance can be obtained using (6) as $Z_0 = 8\ \Omega$. By substituting Z_0 value in (16), the resonant elements values of $L_r = 6.4\ \mu$H and $C_{r1} = 99.5$ nF are calculated. It should be noted that the proposed converter operates at the fixed resonant frequency to provide load-independent output, hence, the switching frequency is replaced by the resonant frequency in (16).

$$L_r = \frac{Z_0}{2\pi f_r}, \quad C_{r1} = \frac{1}{2\pi f_r Z_0}. \tag{16}$$

Increasing C_n, reduces the L_r RMS current and C_{r2} voltage stress, however, it requires a large C_{r2} which negatively affects the converter power density. Hence, the value of C_n is selected equal to 1 and thus, C_{r2} is calculated as $C_{r2} = C_{r1} C_n = 99.5$ nF. The magnetic inductance $L_m \leq 310\ \mu$H is required for fulfilling the soft switching conditions set by (14) and (15) at CV mode, however, reducing L_m leads to increased current stress. Hence, to achieve the switches ZVS turn-ON while avoiding increased conduction loss, $L_m = 200\ \mu$H is selected.

The proposed converter prototype specifications are summarized in Table I. The presented PWM control circuit is illustrated in Fig. 7. According to this diagram, during CC mode, the output current is compared to a reference current using a PI controller. The consequential error signal modifies the duty cycle of the inverter switches, keeping the battery charging current established at 2 A. As the battery pack achieves full charge and the output voltage reaches the threshold of 90 V, the circuit moves into CV mode operation. The battery charger output is controlled at 90 V during CV mode. It should be noted that the auxiliary switch gate control signal is generated by an OR operation between the outputs of the synchronous control circuit and the mode detector comparator part. During CC mode, the outcome of this OR operation turns off the auxiliary switch. Conversely,

TABLE I
IMPLEMENTED BATTERY CHARGER SPECIFICATIONS

Parameter/Component	Symbol	Specification
DC input voltage	V_{in}	340 V
DC output voltage	V_o	50–90 V
Output current	I_o	0.2–2 A
Output power	P_o	18–180 W
CC charge current	I_o	2 A
CV charge voltage	V_o	90 V
Switching frequency	f_s	200 kHz
Resonant inductor	L_r	6.4 μH
Resonant capacitors	C_{r1}, C_{r2}	99.5 nF
Magnetizing inductor	L_m	200 μH
Input capacitors	C_{in1}, C_{in2}	22 μF
Output capacitor	C_o	33 μF
Transformer turns ratio	n	3.75
Main switches	S_1, S_2	LSB60R030HT
Auxiliary switch	S_a	IPP075N15N3G
Secondary diodes	D_{o1}, D_{o2}	SB5100L

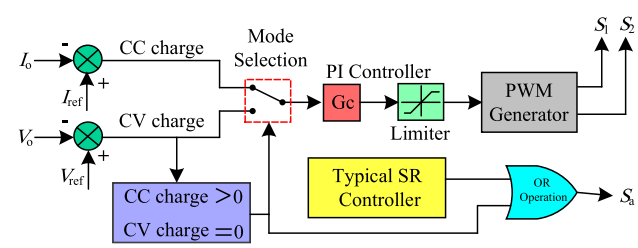

Fig. 7. Control block diagram of the proposed battery charger.

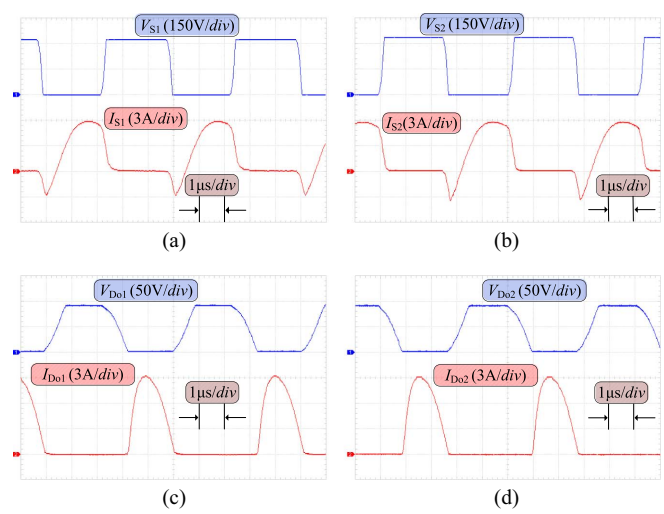

Fig. 8. Experimental voltage and current waveforms for CC mode: (a) S_1; (b) S_2; (c) D_{o1}; and (d) D_{o2}.

in CV mode, the result of this operation applies the synchronous control circuit output to the auxiliary switch gate. The CC and CV modes experimental current and voltage waveforms of the semiconductor devices are depicted in Figs. 8 and 9, respectively. As observed from Fig. 8(a) and 8(b) and Fig. 9(a) and 9(b), the inverter switches ZVS turn-ON are maintained for both operating modes which mitigate switching loss and limits the switches voltage stress to the input voltage. Also, Fig. 8(c) and

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TABLE II
COMPARISON BETWEEN RESONANT BATTERY CHARGERS WITH CC–CV CHARGE

Topologies	[21]	[23]	[28]	[24]	[9]	Proposed
Main switches	4	2	2	2	2	2
Axillary switches	4	0	4	4	2	1
Diodes	6	8	4	4	2	3
Capacitors	2	4	6	3	5	5
Inductors	1	2	4	3	1	1
Transformers	1	2	1	1	1	1
Switching feature in CC state	ZVS	ZVS	ZVS	ZVS	ZVS	ZVS
Switching feature in CV state	ZVS	ZVS	ZVS	ZVS	ZVS	ZVS
Switching frequency range	Medium	Fixed	Fixed	Medium	Fixed	Fixed
Control complexity	Complex	Simple	Simple	Simple	Simple	Simple
Output voltage (V)	60–450	120–160	17–22.5	50–90	50–90	50–90
Rated power (W)	1800	320	50	180	180	180

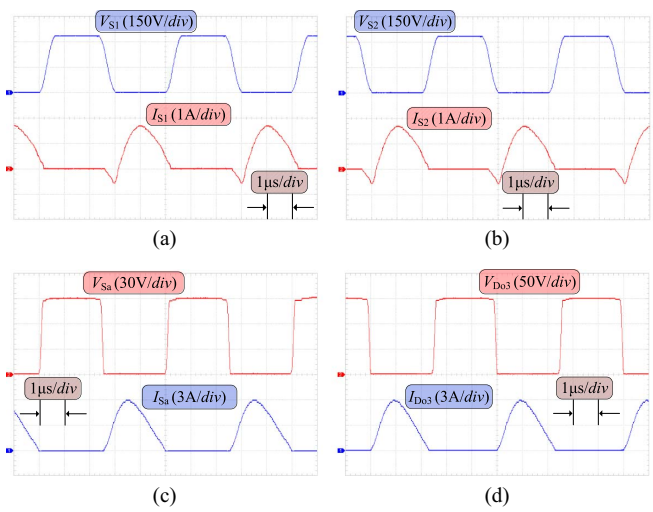

Fig. 9. Experimental voltage and current waveforms for CV mode: (a) S_1; (b) S_2; (c) S_a, and (d) D_{o3}.

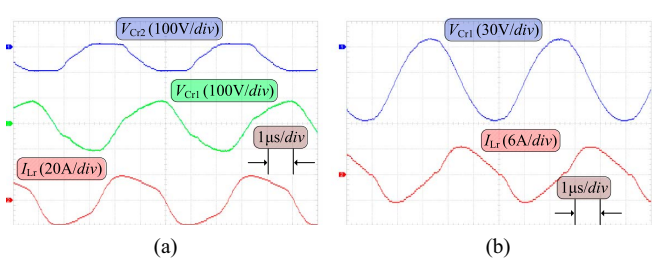

Fig. 10. Experimental waveforms of: (a) I_{Lr}, V_{Cr1}, and V_{Cr2} for CC mode; and (b) I_{Lr} and V_{Cr1} for CV mode.

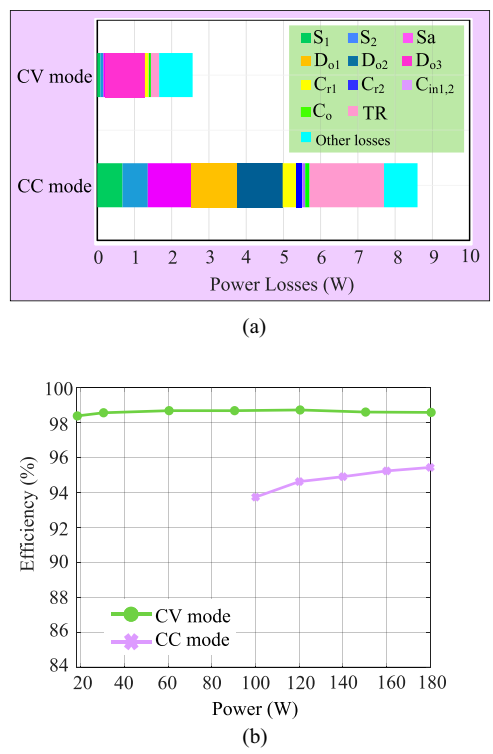

Fig. 11. (a) power loss distribution; and (b) efficiency versus output power for CC and CV modes of proposed battery charger.

8(d) and Fig. 9(c) and 9(d) illustrate that the ZCS turn-OFF is completely provided for rectifier diodes in both CC and CV modes which alleviates the reverse recovery losses. Also, the experimental waveforms of the resonant elements for CC and CV modes are presented in Fig. 10. The proposed converter detailed loss breakdown analysis and the converter efficiency versus output power for CC and CV modes are presented in Fig. 11(a) and 11(b). As seen, the highest proposed converter efficiency is about 98.8% which is resulted from structure simplicity, using synchronous switch in CV mode, operating at the resonant frequency which reduces the circulating power, ZVS operation of the switches and ZCS operation of the diodes.

In Table II, the proposed converter is compared with some RCs offered in [9], [21], [23], [24], and [28] realizing CC–CV charge. The proposed converter has fewer elements compared with other structures and lower control complexity in comparison with [21]. Also, the proposed converter operating frequency range is fixed while [21], [24] are forced to operate at a wide frequency range to realize CC–CV charge. Moreover, the proposed converter utilizes a single auxiliary switch, while in [9], [21], [24], and [28] bidirectional switches realized by two MOSFETs

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are used which increases the converter conduction loss, complexity, and price.

V. Conclusion

A novel simple-structure isolated half-bridge RC is presented in this article with the CC–CV charge capability required for EV batteries without any need for extensive frequency variations common in conventional RCs. The dual resonant network used in this converter alters to LCC network by keeping the auxiliary switch ON continuously to deliver an almost constant output current for fast EV battery charging. While, operating the auxiliary switch as synchronous rectifier reconfigures the series LC network and provides CV battery charging at final charging mode. The proposed battery charger theoretical analysis is verified by the experimental results achieved from a laboratory prototype with wide output voltage range of 50–90 V and wide output current range of 0.2–2 A while operating at the fix resonant frequency. The results indicate the high peak efficiency of 98.8% caused by the converter structure simplicity, using synchronous switch at CV mode, operating at the resonant frequency which reduces the circulating power, ZVS operation of the switches, and ZCS operation of the diodes.

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